

BEM Group Assignment

ROTOR & WAKE AERODYNAMICS

AE4135

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1 Introduction

The performance analysis of wind turbine blades is a critical element of the design process, aiming to maximise energy capture and ensure structural integrity. The method used for aerodynamic performance prediction is the Blade Element Momentum (BEM) theory [3], which combines Froude's momentum theory with the blade element theory developed by Drzewiecki to model the interaction between the rotor blade and the surrounding airflow. This approach is often used for its computational efficiency and reasonable accuracy, making it an excellent choice for preliminary design, despite having some drawbacks as explained in [section 3](#).

In this report, a BEM algorithm has been implemented to analyze a wind turbine rotor with a constant DU airfoil (DU95W180) along the span of the blade. The algorithm is briefly explained in [section 2](#), including the incorporation of aerodynamic corrections to improve the accuracy, especially in conditions where there is a high loading and near the blade tips and roots. Specifically, the Glauert correction is applied to account for the theoretically non-physical behaviour of heavily loaded rotors [3], while Prandtl's tip and root loss correction factors are implemented to model the finite number of blades and the consequential three-dimensional flow effects near the blade ends.

The report begins by investigating a steady case of the BEM model including performance metrics of aerodynamic parameters such as angle of attack, inflow angle, and loading in the spanwise distribution. Furthermore, the total thrust and torque produced by the wind turbine are computed for a range of tip speed ratios to analyse the entire system performance. The power coefficient, which is a valuable indicator of efficiency, is also implemented to indicate the optimal operation conditions for the wind turbine.

A sensitivity study takes place in [subsection 4.2](#) to explore the influence of the tip correction, as well as the number of annuli and the spacing method. This analysis helps evaluate how the tip correction enhances the realism of the results and improves the overall efficiency and accuracy of the code. Further [subsection 4.3](#) addresses the behaviour of the flow around the wind turbine by investigating the distribution of the stagnation pressure and the circulation.

To extend the analysis beyond uniform flow conditions, [section 5](#) focuses on the yawed rotor case by incorporating Glauert's YAW model in an unsteady BEM code. Simulating the effects of rotor misalignment with the wind on the inflow and loading provides valuable performance metrics to understand the rotor's behaviour under more realistic yawed conditions.

With a working BEM code, an optimization aimed at maximizing the power coefficient is showcased in [section 6](#) by adjusting the pitch angle. The new design is presented along with the optimization method. Lastly, the report is concluded by addressing the operational point of the airfoil in terms of lift and drag, along with final remarks.

2 Flowchart

The Blade Element Momentum (BEM) code predicts the aerodynamic performance of a wind turbine by dividing the rotor blade into multiple sections (*annuli*) along the span. At each section, the local forces are calculated based on the airfoil characteristics (lift and drag coefficients), local flow conditions, and induction factors. The code iteratively solves for the axial and azimuthal induction factors a , a' by comparing the forces calculated by blade element theory with those from momentum theory.

These induction factors describe the induced velocities in the axial and azimuthal directions due to the rotor's interaction with the incoming airflow, influencing both the flow angle ϕ and the angle of attack α acting on the blade sections. The angle of attack is used to determine the lift and drag coefficients, C_l , C_d , and calculate the axial and azimuthal force components, C_n , C_t , exerted on the blade section.

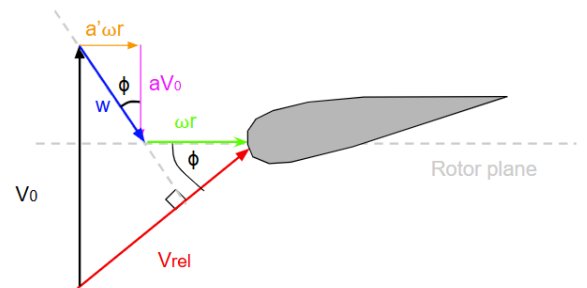


Figure 1: *Velocity triangle at a blade element, showing the effect of induced velocities (a , a') on the relative wind speed and inflow angle.*

To improve accuracy, the previously mentioned correction factors are applied before computing the relative velocity V_{rel} . Finally, the axial and azimuthal loads p_n, p_t acting on the blade for each radial position are calculated before checking for convergence. This process is summarized in Figure 2¹.

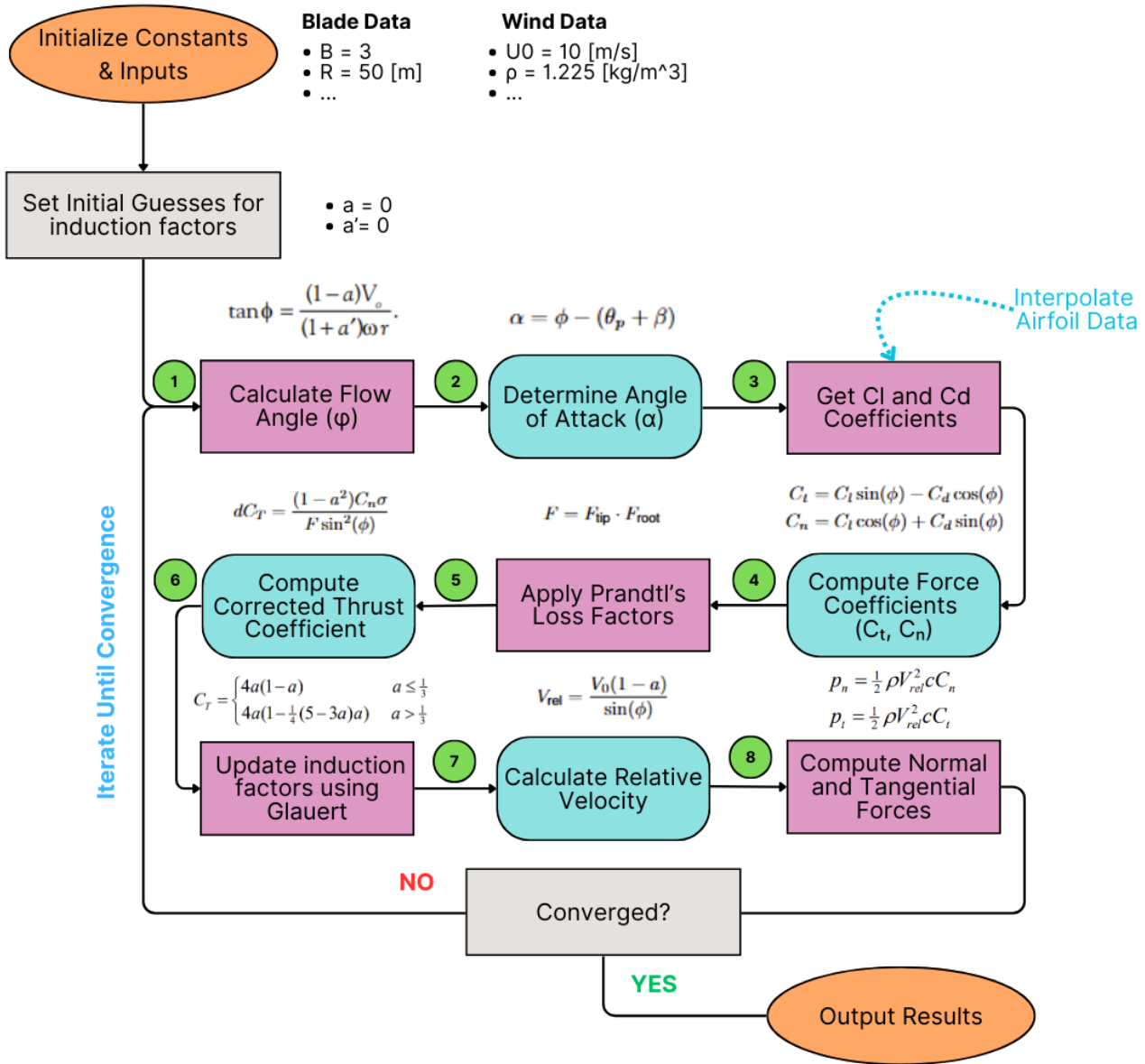


Figure 2: Blade Element Method Algorithm represented in a Flowchart.

3 Main Assumptions

To model the performance of a wind turbine certain assumptions are made in the BEM code. These assumptions can be separated into two categories. Firstly, the assumptions made in the momentum theory and secondly the assumptions related to realism, which simplify the flow case around the wind turbine.

3.1 Assumptions: Modelling rotor

- **Actuator Disk Assumption:** The BEM code treats the rotor blade as a thin actuator disk that extracts momentum from the airflow passing through it, without considering the individual blades. This assumption

¹ In this report, the naming follows [3], where tangential and azimuthal are used interchangeably; symbols are used in the plots: normal load (p_n) = axial load (F_{axial}) and tangential load (p_t) = azimuthal load ($F_{azimuthal}$).

implies that the forces are uniformly distributed across the disk, which disregards certain effects such as blade-vortex interactions and wake dynamics.

- **Small angle approximation:** The small angle approximation is used when implementing the angles and does not hold at large inflow or pitch angles. Results for heavily pitched blades could therefore significantly underestimate drag and overestimate lift, incorrectly predicting power output and stall characteristics.
- **Independent annular streamtubes:** The BEM model assumes purely axial flow through independent annular stream tubes, neglecting radial flow between blade sections. In reality, radial flow redistributes momentum, affecting stall onset, blade loading, and performance. Ignoring this reduces accuracy, particularly in predicting local loads near the blade root and stall behaviour due to centrifugal and Coriolis effects.
- **Airfoil data linear interpolation:** The data for the lift and drag coefficients at each angle of attack is linearly interpolated from the provided polars of the DU airfoil. This relies on the behaviour of the airfoil for the gap between the two closest angles of attack to be linear. This is simply an approximation and therefore leads to inaccuracies.
- **Correction factors:** Correction factors such as Glauert or Prandtl's tip loss factor aim to model the behaviour of the wind turbine based on empirical results. Even though these corrections improve the accuracy of the results, they are still based on empirical data and do not perfectly model the flow conditions.
- **Computational convergence:** The BEM solver uses iterative methods to calculate the induction factors and is therefore subjected to selected convergence criteria to determine the solution. To mitigate this, convergence studies can be performed to make these errors negligible by helping establish appropriate ranges for the variables and margins.
- **Constant airfoil along the span:** This BEM code assumes a single airfoil for the whole blade, but real wind turbine blades employ multiple different airfoils. They typically use thicker designs near the root for increased structural robustness and progress towards more aerodynamically efficient airfoil designs as the span moves towards the blade tip. By only having one airfoil, structural and aerodynamic requirements must be balanced, which particularly compromises performance at the blade tip regions and even a potential early stall.

3.2 Assumptions: Operating conditions

The BEM code assumes ideal inflow conditions, to simplify the mathematical modelling. Ideal inflow conditions imply a uniform inflow, which is inviscid, incompressible and steady in time. These conditions are closer to laminar flow conditions and ignore effects such as transient phenomena, gusts, turbulence, wind shear, and inflow angle variations which are often present in real wind turbines.

- **Uniform inflow:** is assumed meaning the flow velocity is constant $U_0 = 10[\text{m/s}]$ across the entire swept area A . This is often not true in real life due to variations in wind speed and direction caused by turbulence, terrain effects, and atmospheric boundary layers. However, the BEM code could be extended to account for dynamic inflow conditions in future studies [1].
- **Inviscid:** viscosity is disregarded, such that boundary layer separation and stall are not modelled and are inferred indirectly through the empirical airfoil data. In real life the presence of viscous leads to boundary layer formation, separation, and potential stall, especially at higher angles of attack. The BEM code can be expanded using the Vortex Panel Method to account for viscosity [1].
- **Incompressible:** means that the wind turbines operate below 0.3 Mach, which is often the case in real life for typical design tip-speed ratios to avoid compressibility effects.
- **Steady:** Although unsteady BEM codes exist to model certain effects caused by the rotation, such as the wake and vortex shedding. In this report, the steady BEM code is used except for the YAW model.

The BEM code does not account for any structural deformation which is caused by the exerted forces. When the air exerts a force on rotor blades they bend due to elastic effects and result in different aerodynamic properties by changing the angle of attack. These phenomena are disregarded in the BEM code. In addition, the tower shadow which is the influence of the wind turbine tower on the flow is disregarded in the BEM code.

To model more realistic turbulent wind conditions, as well as the flexibility effects of the rotor blades, CFD including turbulence modelling and fluid-structure interaction must be implemented.

4 Steady BEM Algorithm

The steady BEM algorithm is implemented as described in [section 2](#) to investigate the performance across the tip speed ratios $\lambda = 6, 8, 10$ and for a wind speed of $V_0 = 10$ [m/s]. The same airfoil of *DU 95-W-180* is used ranging from $r/R = 0.2 - 1.0$.

4.1 Performance Metrics

The efficiency of the wind turbine is captured by the power coefficient C_P , which is a ratio of kinetic energy from the wind to mechanical power, as given in [Equation 1](#), where $A = 2\pi R$:

$$C_P = \frac{P}{\frac{1}{2}\rho AV_0^3} \quad (1)$$

The power P depends on the torque exerted on the rotor by the incoming wind and the rotor's angular velocity ω , as given by [Equation 3](#):

$$M_r = B \int r p_t dr \quad (2)$$

$$P = \omega M_r \quad (3)$$

The performance of the wind turbine is summarized in [Figure 3](#), where the power coefficient C_P is plotted for different tip speed ratios λ . The power coefficient is seen to be the highest at a tip speed ratio of $\lambda \approx 9$.

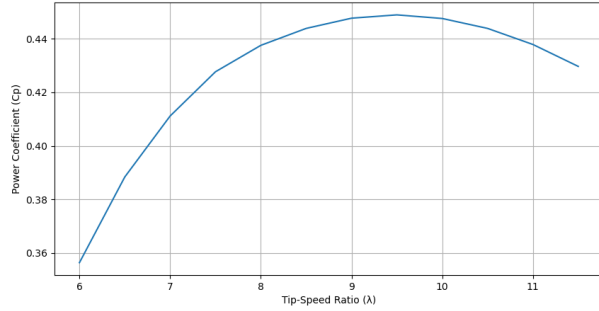


Figure 3: Power coefficient C_P plot against tip speed ratio λ

4.1.1 Spanwise distribution of angle of attack and flow angle

Looking at the velocity triangle shown in [Figure 4](#) it the definition of the flow angle ϕ can be derived to [Equation 4](#).

$$\tan(\alpha) = \frac{(1-a)V_0}{(1+a')\omega r} \quad (4) \quad u = V(1-a)$$

Furthermore, the angle of attack α is given by [Equation 5](#) in terms of the flow angle ϕ .

$$\alpha = \phi - \theta$$

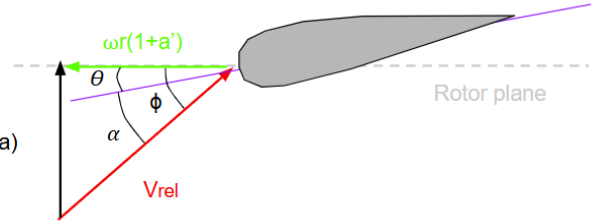


Figure 4: Velocity triangle showing the velocity components acting on the rotor blade section

From these two equations, it is expected that the flow angle is large near the root where the radial distance r is small and decreases along the span until the tip. The same behaviour can be expected for the angle of attack because it is dependent on the flow angle, but lower due to the airfoil pitch θ . Consider the results shown in [Figure 5](#).

The resulting angle behaviour is as expected, but also includes a dip near the root and tip of the airfoil. This is due to Prandtl's root and tip loss correction factors F_{root} and F_{tip} , which are explored in more detail in [subsection 4.2.1](#). Furthermore, the angles are seen to decrease with the tip speed ratio λ . Considering the definition of the tip speed ratio $\lambda = \frac{\omega r}{V_0}$ and the flow angle it can be said that $\tan(\phi) \propto \frac{1}{\lambda}$ the angles to decrease with larger tip speed ratios.

An important observation can be made about the angle of attack for $\lambda = 6$, which is the only one that exceeds 10 degrees. At angles of attack larger than 10 degrees the blade starts to stall, as mentioned in [subsection 6.2](#). This causes unfavorable aerodynamics which will be addressed in the upcoming sections.

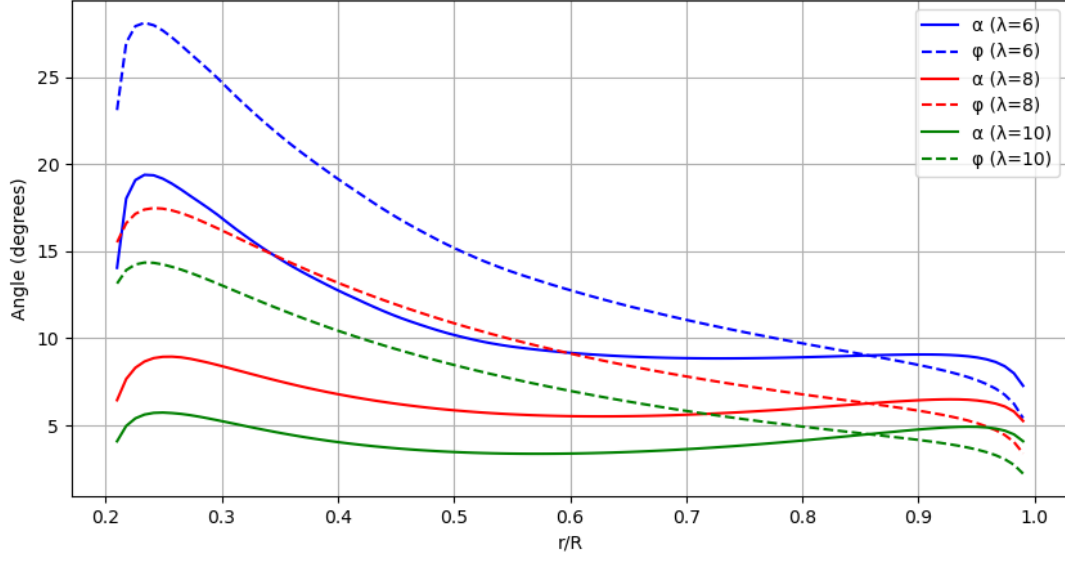


Figure 5: Spanwise distribution of angle of attack α and inflow angle ϕ for tip speed ratio $\lambda = 6, 8, 10$

4.1.2 Spanwise distribution of induction factors

One-dimensional momentum theory for an ideal wind turbine defines the induction factors in Equation 6 and Equation 7. Based on these equations it can be said that the axial induction factor a is dependent on how fast the wind is moving through the rotor disk, while the azimuthal induction factor a' should decrease with $\frac{1}{r}$ and tend to zero.

$$u(r) = (1 - a)V_0 \quad (6)$$

$$V_0 = (1 + a')\omega r \quad (7)$$

However, the BEM code features correction factors including Glauert and Prandtl's tip and root loss, which adjust the induction factors. For this reason, the axial induction factor experiences an increase at the root and tip. This is illustrated in Figure 6.

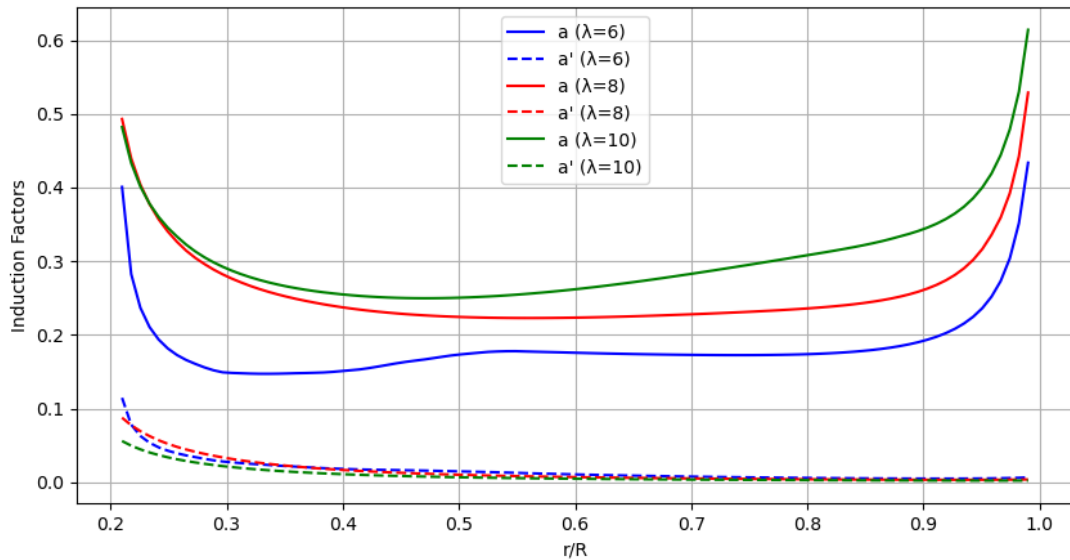


Figure 6: Spanwise distribution of axial and azimuthal induction factors a, a' for tip speed ratio $\lambda = 6, 8, 10$

The resulting induction factor a is seen to gradually decrease towards the centre of the blade until it reaches a plateau and starts to increase again. The induction factor can also be defined in terms of the thrust $C_T = 4a(1 - a)$ which increases with distance along the blade explaining the steady rise of axial induction a . In addition, the stall for $\lambda = 6$ can be seen near the root in the airfoil blade.

4.1.3 Spanwise distribution of axial and azimuthal loads

The loads experienced by the blade per unit span are given by Equation 9, where is taken from the velocity triangle to be $V_{rel} = \sqrt{(V_0(1 - a))^2 + (\omega r(1 + a'))^2}$.

$$p_n = \frac{1}{2} \rho V_{rel}^2 c \underbrace{C_l \cos(\phi) + C_d \sin(\phi)}_{C_n} \quad (8)$$

$$p_t = \frac{1}{2} \rho V_{rel}^2 c \underbrace{C_l \sin(\phi) - C_d \cos(\phi)}_{C_t} \quad (9)$$

Figure 7 illustrates the variation of the loads along the span. The axial load is seen to increase with the radius of the blade and peaks near the tip at roughly $r/R \approx 0.95$. This load increases with the tip speed ratio λ and the stall effects seen at $\lambda = 6$ are noticeable by the decrease near the root. On the other hand, the tangential load is seen to remain at a constant value throughout the mid-span of the blade and decrease and the tip and root.

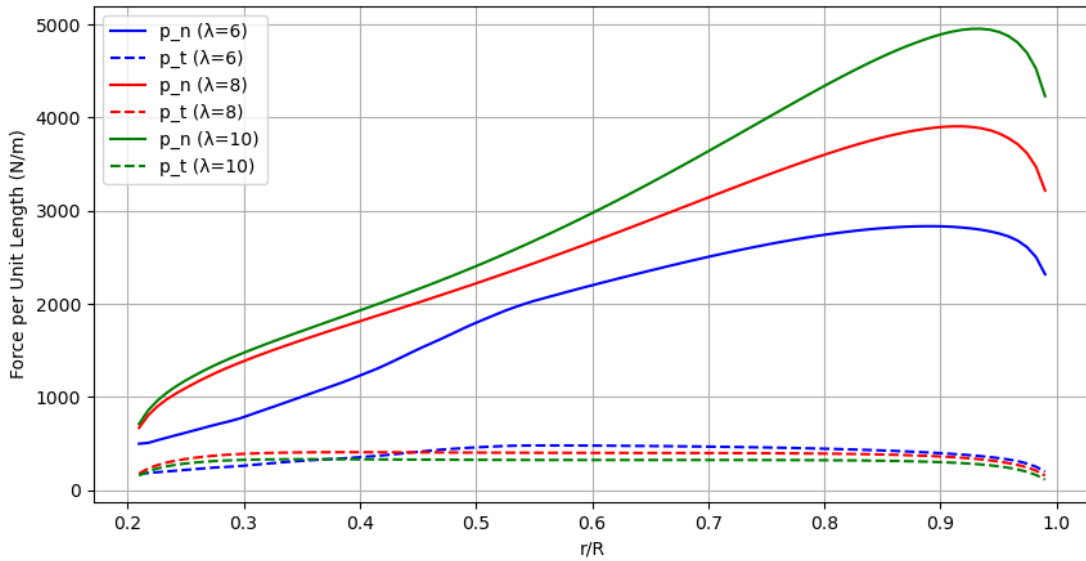


Figure 7: Spanwise distribution of thrust and azimuthal loading p_n, p_t for tip speed ratio $\lambda = 6, 8, 10$.

4.1.4 Total thrust and torque

To calculate the total thrust and torque produced by the wind turbine the loads are integrated over the blade length as shown by Equation 11.

$$M_r = B \int r p_t dr \quad (10)$$

$$T = B \int p_n dr \quad (11)$$

The results can be seen in Figure 8, showing that the torque M_r peaks at roughly $\lambda = 6.5$ and then decreases with a larger tip speed ratio. On the other hand, the thrust increases with the tip speed ratio.

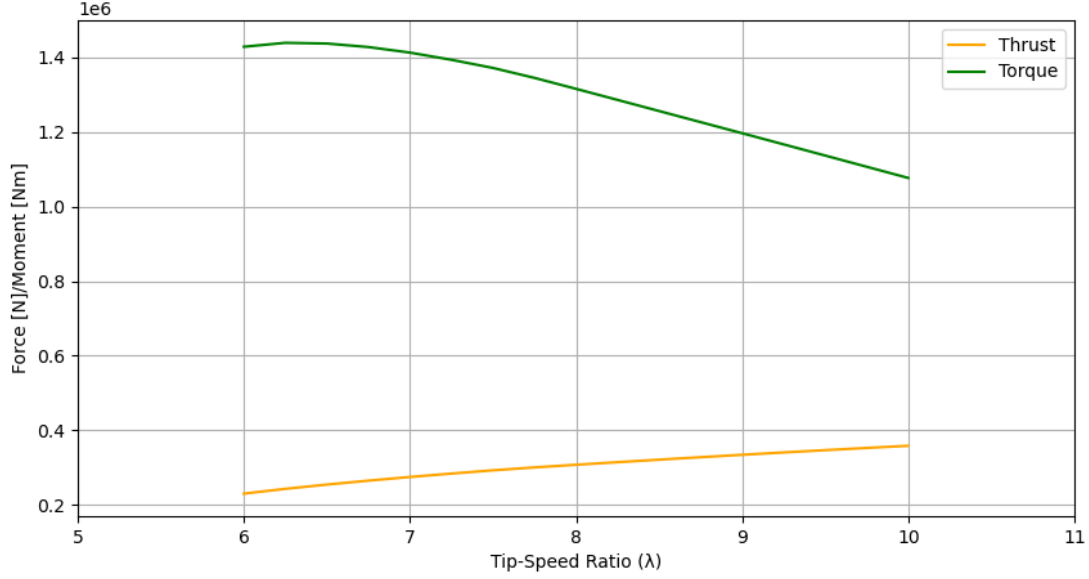


Figure 8: Total thrust and torque for tip speed ratio range $\lambda = 6 - 10$.

4.2 Sensitivity Studies

4.2.1 Influence of tip correction

Prandtl's tip loss correction factor attempts to account for the 3D aerodynamic effects at the tip of the rotor blades. This is not captured by the 2D momentum theory because an infinite number of blades is assumed. Hence equation Equation 12 is used for 3 blades.

$$F_{\text{tip}} = \begin{cases} \frac{2}{\pi} \cos^{-1} \left(\exp \left(-\frac{B}{2} \cdot \frac{R-r}{r \cdot \sin |\phi|} \right) \right), & \text{if tip correction is applied} \\ 1, & \text{otherwise} \end{cases} \quad (12)$$

Figure 9 illustrates the influence of Prandtl's tip loss correction factor. It can be seen that without the correction the load continues to increase near the tip. In reality, the imbalance of the pressure across the upper and lower surface starts to decrease due to 3D spill-over effects causing tip vortices. These vortices reduce the effective angle of attack and induce a downwash which ultimately causes the lift to decrease. The same effect is seen to take place in the azimuthal load distribution shown in Figure 10.

The tip loss correction factor is seen to take effect at approximately $r/R \approx 0.8$ for the axial load p_n and slightly closer to the root at roughly $r/R \approx 0.7$ for the azimuthal load p_t . Without the tip correction factor, the axial force per unit length is incorrectly higher by roughly 1200 [N/m] and the azimuthal load by approximately 150 [N/m]. These differences highlight the importance of this correction factor without which the output power would be significantly overestimated.

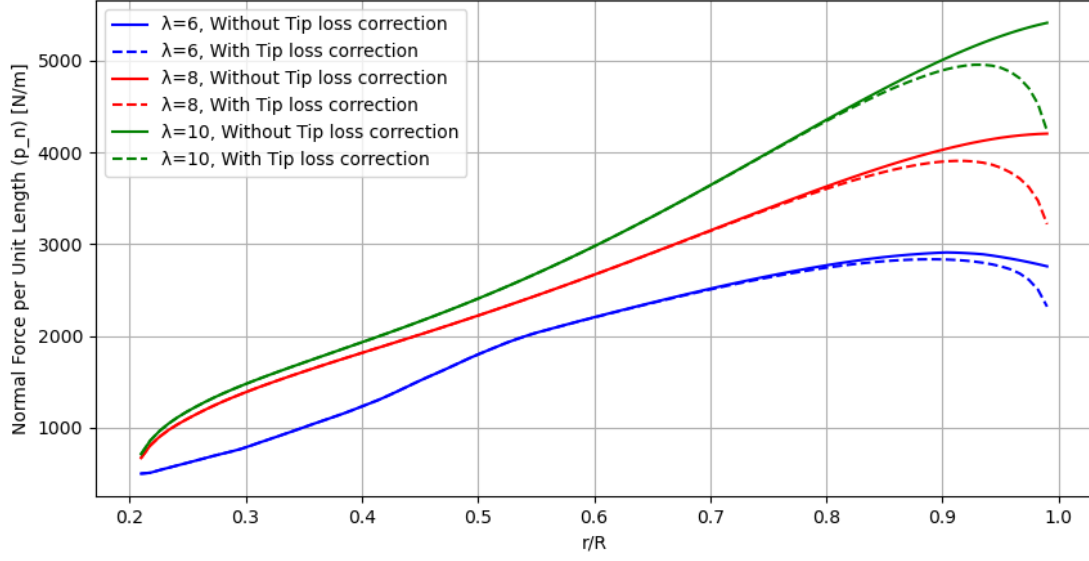


Figure 9: Spanwise distribution of axial load p_n with and without Prandtl's tip correction for $\lambda = 6, 8, 10$.

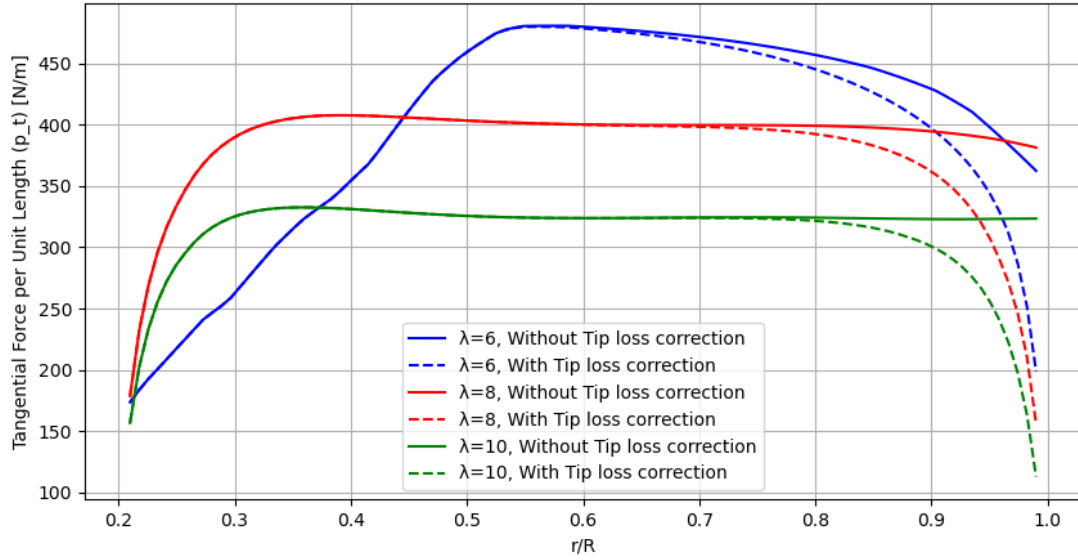


Figure 10: Spanwise distribution of azimuthal load p_t with and without Prandtl's tip correction for $\lambda = 6, 8, 10$.

4.2.2 Influence of number of annuli and spacing method

The effect of node number and spacing on thrust convergence can be analyzed to determine the optimal discretization for accurate force calculations. A sufficient number of nodes is necessary to capture radial variations in aerodynamic forces, but this should be done without computational efficiency. Figure 11 illustrates how different node counts and spacing methods, namely uniform and cosine spacing, influence the convergence of total thrust.

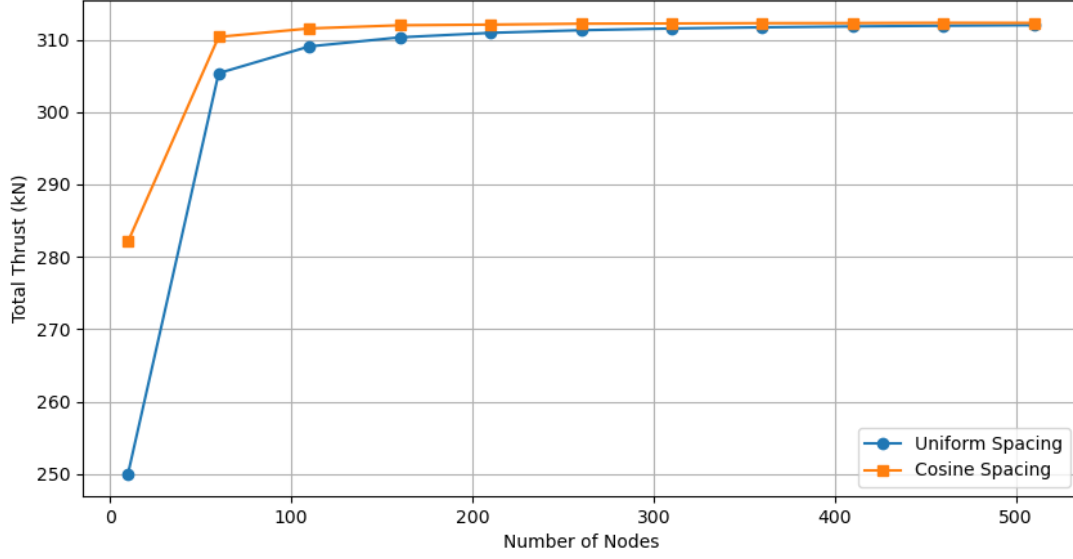


Figure 11: Effect on node number and spacing on the convergence of total thrust

Figure 11 shows that as the number of nodes/annuli increases, the total thrust converges to a stable value of approximately 313kN. For lower node counts (e.g. below 100), the thrust values differ considerably from the converged final value, particularly for uniform spacing. This suggests that the discretization is too coarse, which leads to inaccuracies in the force calculations, as there is insufficient resolution to capture radial variations in aerodynamic forces. However, beyond approximately 150 nodes, the thrust stabilizes, indicating that further increases in node count provide minimal additional accuracy while increasing computational time considerably.

Furthermore, the choice of spacing method, and thus node distribution, significantly affects the convergence behaviour. The cosine spacing method reaches the asymptotic thrust value much more rapidly than the uniform spacing method, which requires a higher number of nodes to achieve the same level of convergence. This is because the cosine spacing places more nodes near the blade tip where the normal forces are much higher. Therefore, accurately modelling this region, which contributes the most to the thrust, results in a more precise overall thrust calculation for the entire blade. This demonstrates that a non-uniform node distribution can significantly improve computational efficiency without sacrificing accuracy.

4.3 Flowfield Characterization: Stagnation Pressure and Circulation Effects

4.3.1 Stagnation pressure distribution

Stagnation pressure (also called total pressure) is the sum of the static pressure and dynamic pressure in a fluid flow. It represents the pressure a fluid would have if it were brought to rest isentropically. In equation form, this translates to:

$$p_{stag} = p_{static} + \frac{1}{2}\rho V^2 \quad (13)$$

The change of pressure across an actuator disk exerting an axial force f_z on the flow can be calculated using Equation 14, where the axial force is defined in Equation 15. Assuming that the force acting on the disk is constant over the surface and only has a component in the z-direction (i.e. normal to the disk), as well as assuming that the vorticity is zero at this point, this equation can be further simplified as shown:

$$\vec{\nabla} h_s = \begin{bmatrix} \frac{\partial h_s}{\partial r} \\ \frac{\partial h_s}{\partial \theta} \\ \frac{\partial h_s}{\partial z} \end{bmatrix} = \begin{bmatrix} v_\theta \omega_z - v_z \omega_\theta \\ -v_r \omega_z + v_z \omega_r \\ v_r \omega_\theta - v_\theta \omega_r \end{bmatrix} + \frac{1}{\rho} \begin{bmatrix} f_r \\ f_\theta \\ f_z \end{bmatrix} = \frac{1}{\rho} \begin{bmatrix} 0 \\ 0 \\ f_z \end{bmatrix} \quad (14)$$

$$f_z = \frac{B p_n(r) dr}{2\pi r dr} \quad (15)$$

Assuming that the flow only sees a pressure drop at the turbine, this can be plotted as a function of radius for points infinitely before and after the rotor, as well as immediately before and after the rotor. This is presented in

Figure 12a, which shows that a smaller blade radius will create a larger pressure drop, as the area over which the thrust acts is much smaller. Note that only the pressure drop due to the normal force is being considered, meaning that the pressure losses due to tangential forces in the wake are being neglected. Moreover, the stagnation pressure before and after the rotor can be plotted as a function of blade span as shown in Figure 12b. The graph shows an almost constant drop in pressure across most of the span after the rotor, with a lower drop near the tip and the root due to the applied correction factors.

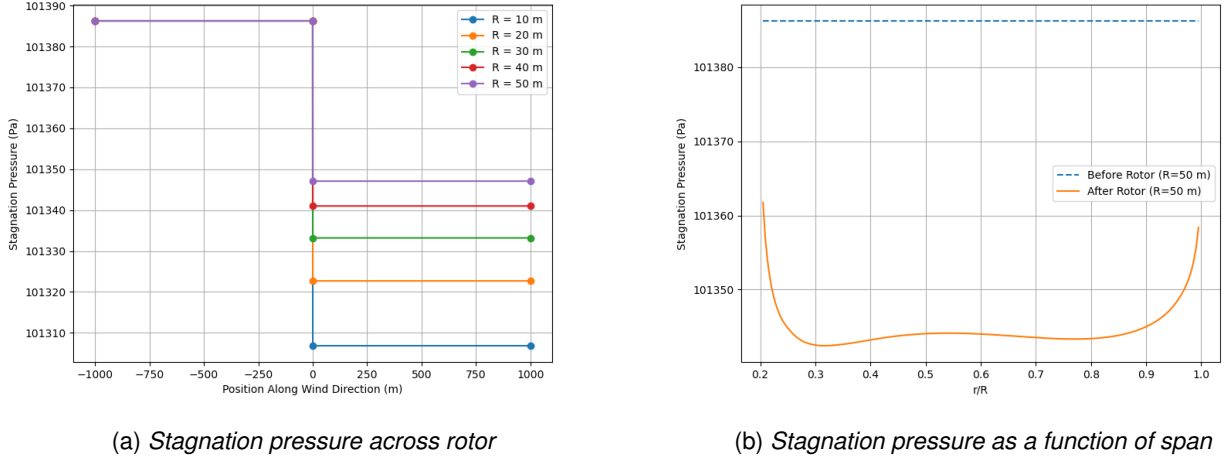


Figure 12: Stagnation pressure distributions across rotor and along span

4.3.2 Circulation in spanwise direction

Circulation at a given blade element is a measure of the rotating flow known as vorticity around the airfoil section. The definition of circulation Γ is given by the Kutta-Joukowski theorem in Equation 16.

$$L = \rho V_{rel} \Gamma \quad (16)$$

To calculate the distribution of circulation along the blade element, the lift per unit span $L' = \frac{1}{2} \rho V_{rel}^2 C_l c$ is used to rewrite Equation 16 into the following:

$$\Gamma = \frac{1}{2} V_{rel} C_l c \quad (17)$$

Implementing Equation 17 in the BEM code yields the distribution of circulation along the blade length, illustrated in Figure 13. The key observation is that circulation increases with the tips speed ratio λ . The general behaviour of the circulation is an increase at the root $r/R \approx 0.1$ where there is a lower rotational speed and less efficient lift coefficient from the angle of attack. In addition, the same effect can be seen at the tip of the blade $r/R \approx 0.9$ where tip losses caused by pressure spilling to the suction side reduce lift. These effects are captured by Prandtl's loss factors.

Furthermore, the general behaviour of the circulation is the same across different λ , except for the circulation near the root at $\lambda = 6$. This is likely due to the stall behavior explained beforehand.

Vorticity is related to circulation by Helmholtz's vortex theorems, which state that vorticity must be conserved leading to trailing vortices. The strength and distribution of vorticity shed into the wake are directly related to circulation meaning.

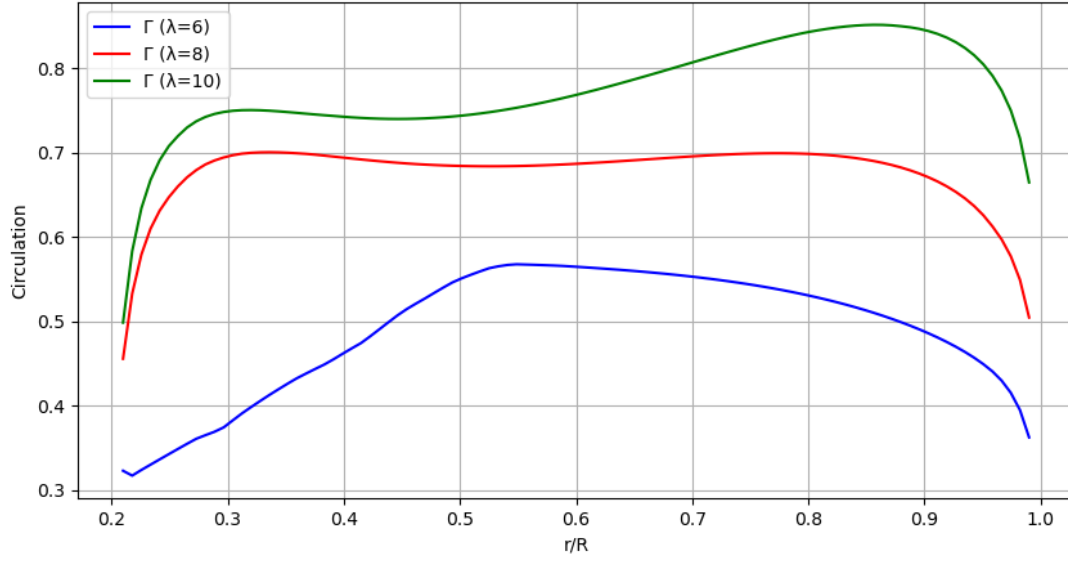


Figure 13: Spanwise distribution of circulation Γ for tip speed ratio $\lambda = 6, 8, 10$.

5 Unsteady BEM: Yawed model

5.1 Introduction

The yaw angle θ_{yaw} is the wind turbine's angle with respect to the incoming wind as shown in Figure 14. Due to this angle, the incoming flow is not aligned with the rotor axis, meaning the aerodynamics of the blade is dependent on the azimuthal angle ψ . In a yawed configuration, the flow is no longer axisymmetric, which causes the wake behind the rotor to develop around an axis different from the rotor axis, called the wake axis. The angle between the rotor axis $\underline{e}_n$ and the wake axis is given as the wake skew angle χ , as illustrated in Figure 14.

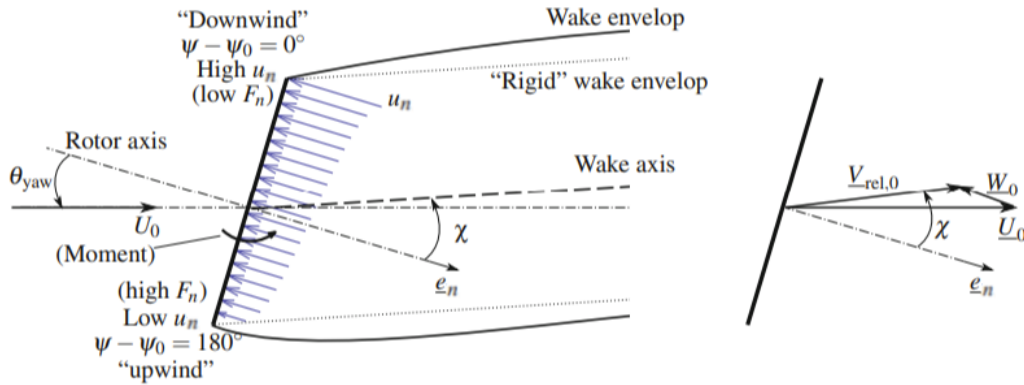


Figure 14: Yawed configuration and resulting velocity triangle. Adapted from [1]

The wake skew angle can be expressed using an induced velocity W_0 illustrated in the velocity triangle Figure 14. Writing the velocity triangle as $V_{rel,0} = U_0 + W_0$. The induced velocities in the yawed configuration differ from the aligned case along the azimuthal angle since the blade may be up or downwind. The part of the rotor that is upwind is expected to have lower axial induction and lead to higher normal loads (below stall). On the other hand, the downwind position will have more induction and result in lower normal loads (also below stall).

5.2 Glauert Yaw model

Implementing a yaw model in the BEM code requires another iteration loop around the azimuthal angle ψ to account for the different blade positions. Furthermore, the wake skew angle χ must be determined to calculate the additional velocity that is induced W_0 due to the yaw angle. χ may be determined using empirical relations such as the one suggested by [2] shown in Equation 18, where $\bar{a}$ is the average induction factor taken at $r/R \approx 0.7$ [1].

$$\chi = \theta_{yaw}(1 - 0.6\bar{a}) \quad (18)$$

This Glauert yaw correction model is based on a lifting line analysis, including the assumptions needed and the following assumptions:

- Induced velocity is constant along the wing chord
- The variation between the mean induced velocity and rotor loading is assumed to be proportional to the distance behind the centre of the rotor
- Induced velocity normal to the rotor U_n can be decomposed into a component that changes linearly with the radius and sinusoidally with the azimuth angle and a constant component.

The Glauert yaw correction model is shown in Equation 19, where $u_{n,0}$ is the induction of the aligned case, ψ is the azimuthal position of the blade and ψ_0 is the reference position set to $-\pi/2$ such that the downwind part of the rotor is on the right-hand side when looking at the wind turbine head on.

$$u_n = u_{n,0} \left[1 + \frac{r}{R} \tan\left(\frac{\chi}{2}\right) \cos(\psi - \psi_0) \right] \quad (19)$$

This correction is implemented in the code before computing every flow angle ϕ , which depends on the normal velocity component as shown in Figure 2.

5.3 Performance results

The unsteady BEM code implementing the correction mentioned above has been used to determine the performance of the wind turbine at tip speed ratio $\lambda = 8$ and three yaw angles $\theta_{yaw} = 0^\circ, 15^\circ$ and 30° . Figure 15 indicates that the power coefficient C_P decreases with the yaw angle, which suggests that the optimal scenario is when a wind turbine is aligned with the flow case. Based on momentum theory, the effective wind speed is decreased by the yaw angle, leading to losses in the wake and a decrease in the power output. This is why modern wind turbines feature yaw control that aligns the wind turbine with the wind direction.

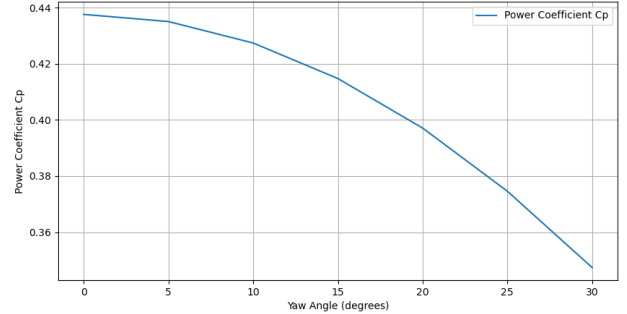


Figure 15: Power coefficient C_P for a range of yaw angles θ_{yaw} .

Looking at the variation of trust and torque illustrated in Figure 16 it can be seen that the torque experiences a significant drop with the yaw angle. The total torque drops by approximately 0.4×10^6 [Nm], which is not ideal for a wind turbine aiming to generate power. On the other hand, the thrust is also seen to decrease, which is an appealing result based on a structural perspective.

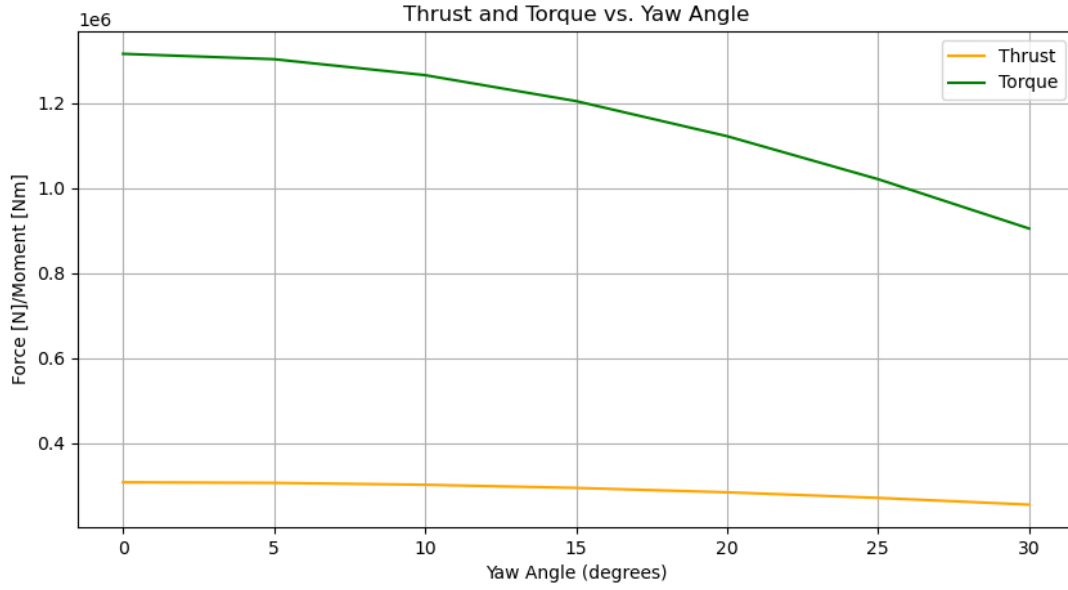


Figure 16: Total thrust and torque variation with yaw angle θ_{yaw} over a range of 0° to 30° .

The following contour plot [Figure 17](#) shows the angle of attack α and the flow angle ϕ in the spanwise distribution and the azimuthal angle. As can be seen in the case of an aligned wind turbine, the results are independent of the azimuthal angle. The highest angle of attack is near the root and decreases with the rotor blade length expected from [Figure 5](#). For the yawed cases, the angle of $\theta_{yaw} = 15^\circ$ and 30° is shown to have a higher angle of attack on the blades upwind than downwind. At the azimuthal angle of $\psi = 0^\circ$ and 270° corresponds to the non-yawed case.

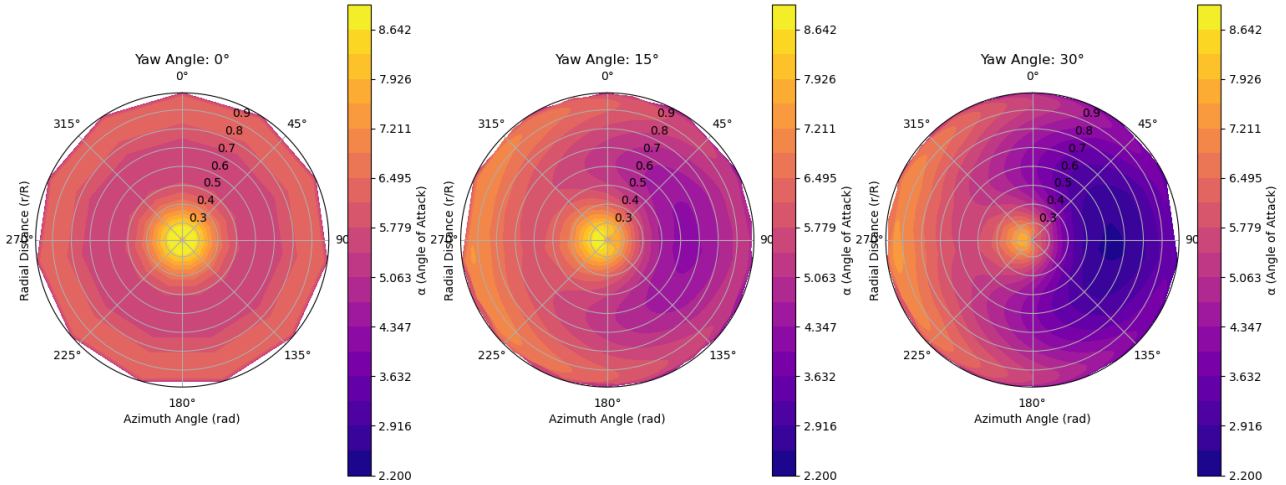


Figure 17: Contour plot showing variation of angle of attack α along the spanwise distribution for the three yaw angles $\theta_{yaw} = 0^\circ, 15^\circ, 30^\circ$.

[Figure 18](#) indicates the change with flow angle ϕ showing a similar behaviour as the angle of attack.

Looking at the loads experienced by the rotors p_n and p_t shown in [Figure 19](#) and [Figure 20](#), it can once again be seen that the non-yawed case is independent of the azimuthal angle. The normal load peaks close to the tip of the rotor blade and decreases towards the root of the blade as expected from the non-yawed case. Looking at the influence of the yaw angle, it can be seen that the yaw angle has a higher axial load at $\theta_{yaw} = 30^\circ$ than $\theta_{yaw} = 15^\circ$ in the upwind section. This behaviour is also seen in the azimuthal load case, except in a less circular fashion.

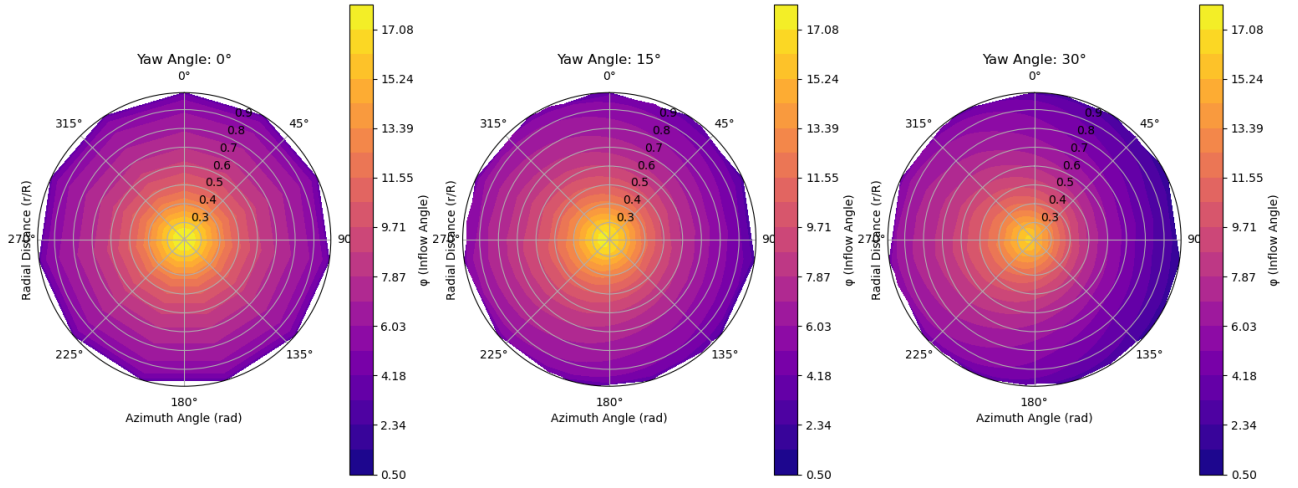


Figure 18: Contour plot showing variation of flow angle ϕ along the spanwise distribution for the three yaw angles $\theta_{yaw} = 0^\circ, 15^\circ, 30^\circ$.

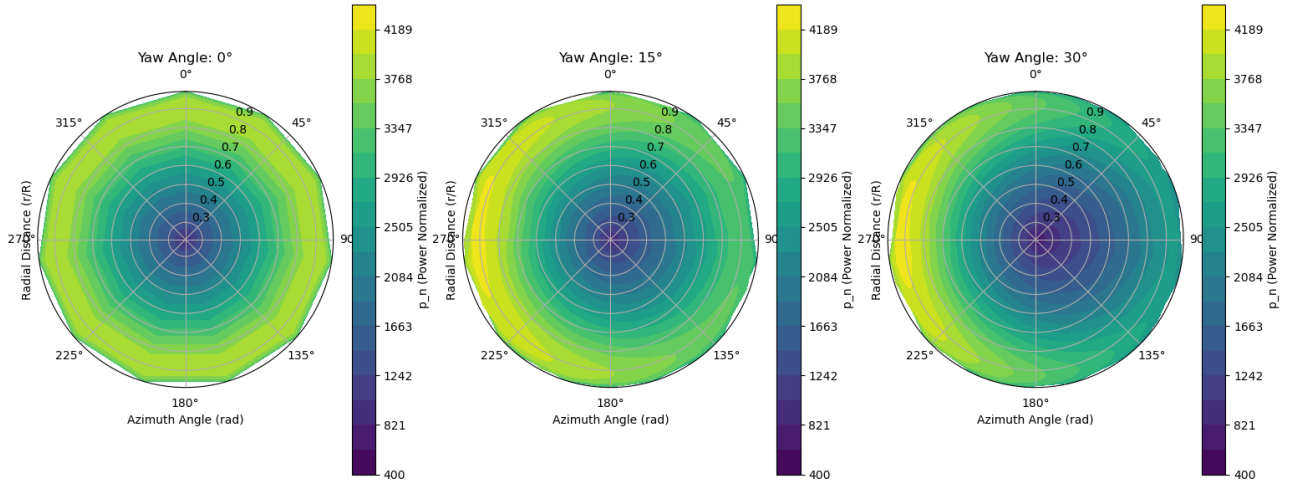


Figure 19: Contour plot showing variation of normal load p_n along the spanwise distribution for the three yaw angles $\theta_{yaw} = 0^\circ, 15^\circ, 30^\circ$.

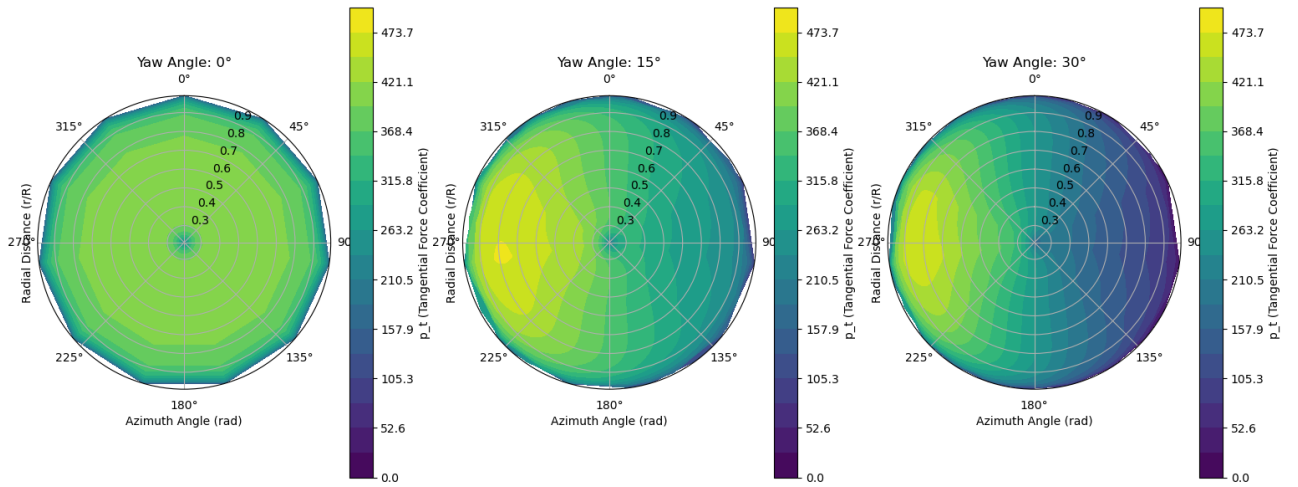


Figure 20: Contour plot showing variation of azimuthal load p_t along the spanwise distribution for the three yaw angles $\theta_{yaw} = 0^\circ, 15^\circ, 30^\circ$.

6 Performance Optimization

6.1 Design for maximum C_P

The design approach taken to optimize the power coefficient, and thus power extraction from the flow, is to modify the pitch angle of the blades. By changing the pitch angle, the angle of attack experienced by the blades is adjusted, which in turn affects the lift and drag forces acting on them. An optimal pitch angle ensures that the lift-to-drag ratio is maximized, leading to higher aerodynamic efficiency and improved power output. This effect could be further improved by changing the tip-speed ratio of the turbine find the most optimal combination to produce the highest C_P possible. This approach results in the contour plot shown in Figure 22, which yields a maximum C_P value of 0.472 at a tip-speed ratio of 8.5 and pitch angle of -5° .

However, the assignment specifies that the C_P optimization is constrained to a fixed value for the tip-speed ratio of 8, as well as requiring an output of thrust coefficient $C_T = 0.75$ for the turbine. Consequently, a contour plot showing the variation of C_T with tip-speed ratio and pitch angle can be created, as shown in Figure 21. This plot features a line for which the condition of $C_T = 0.75$ is met, and thus the point where this line crosses tip-speed ratio = 8 is where the optimized pitch angle lies. This approach yields a pitch angle of -4.25° , which corresponds to a C_P value of 0.466 in Figure 22. This is consistent with the actuator disk model theory, which predicts a maximum theoretical C_P value of $16/27 = 0.593$, as determined by the Betz limit.

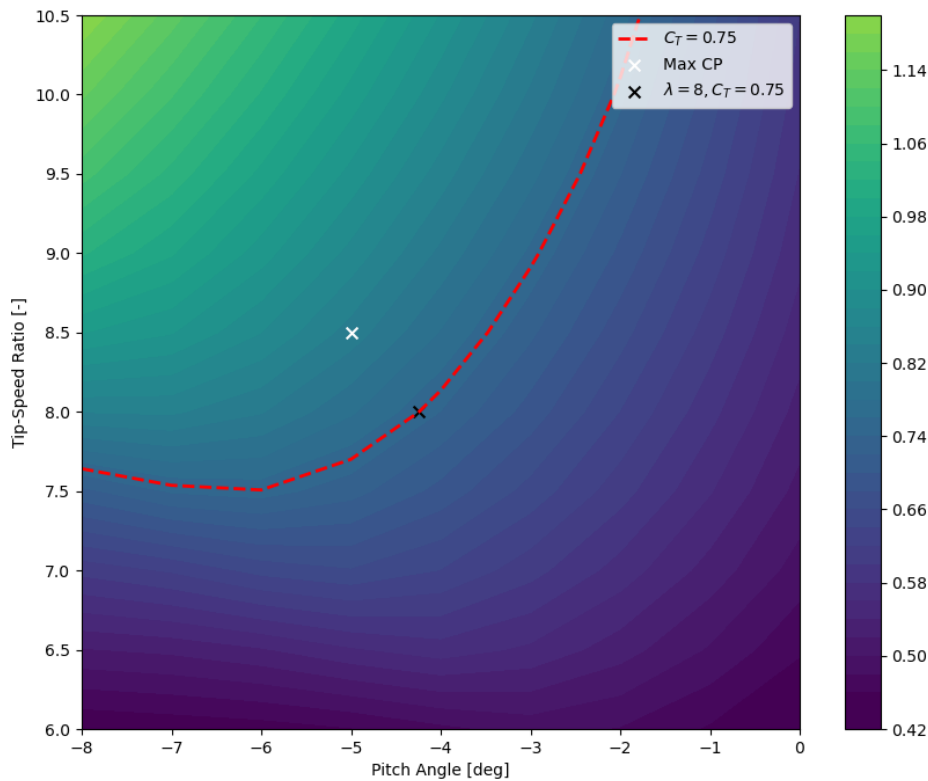


Figure 21: Thrust coefficient, C_T , contour plot for variations in TSR and pitch angle

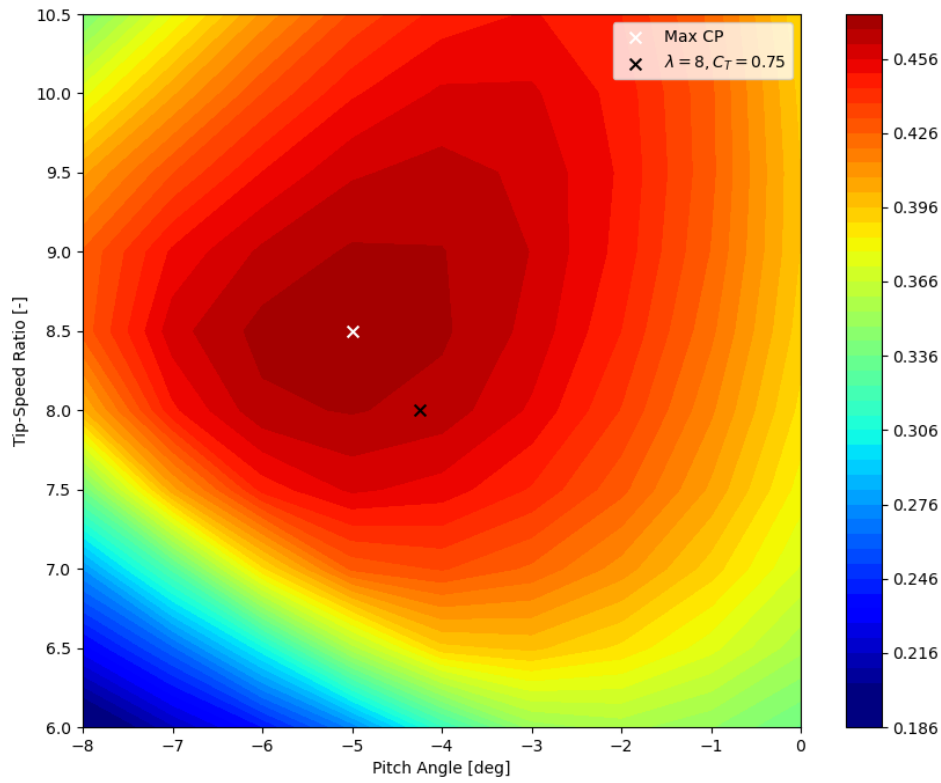


Figure 22: Power coefficient, C_P , contour plot for variations in TSR and pitch angle

The power coefficient for the new pitch angle can also be compared to the original power coefficient curve, as shown in Figure 23. This graph illustrates that the new pitch angle results in a higher peak C_P , thus improving the turbine's efficiency. In fact, the C_P values within the desired tip-speed ratio range are consistently higher than those for the original design, which had a fixed pitch angle of -2° . However, it is important to note that the newly determined optimal pitch angle is only ideal for the specific wind speed being evaluated and will vary depending on the current atmospheric conditions.

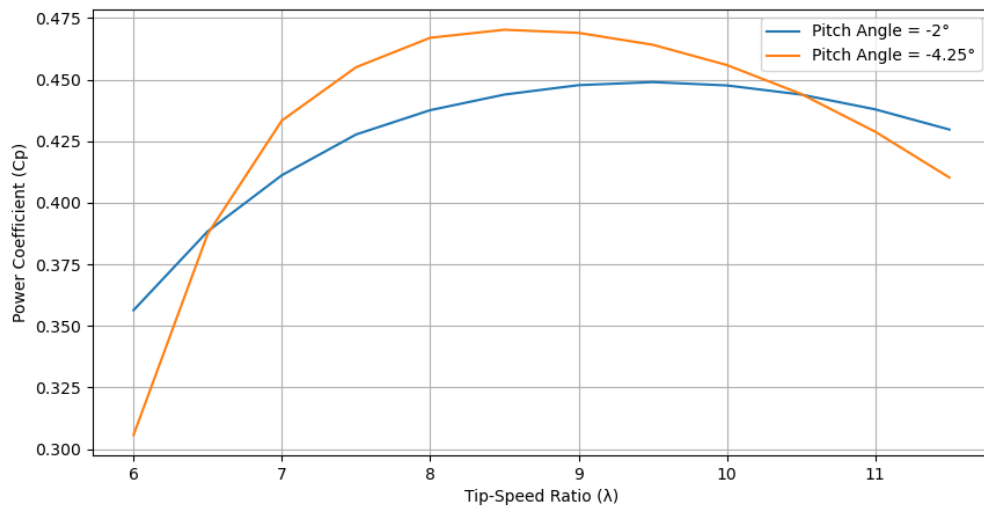


Figure 23: Comparison of C_P for range of tips speed ratio λ for optimized and initial design

6.2 Lift and Drag Analysis

6.2.1 Operational Point of the Airfoil

The operational point of a wind turbine airfoil is determined by analyzing its aerodynamic performance, specifically the variation of lift and drag characteristics with respect to the angle of attack. Figure 24a shows how the lift coefficient (C_l) for the chosen airfoil increases linearly with the angle of attack (α) up to a critical stall angle, beyond which the airfoil stalls due to flow separation. This causes a sharp drop in lift, making it essential to operate below this limit for stable and efficient energy extraction. As mentioned in previous sections of the report, for a tip speed ratio of $\lambda = 6$, the angle of attack exceeds $\alpha \approx 10^\circ$ and therefore causes stall decreasing the lift. Furthermore, Figure 24b illustrates the relationship between C_l and the drag coefficient (C_d), showing that while higher lift can be achieved at larger angles of attack, it comes at the cost of increased drag, which negatively impacts the efficiency of the turbine blade.

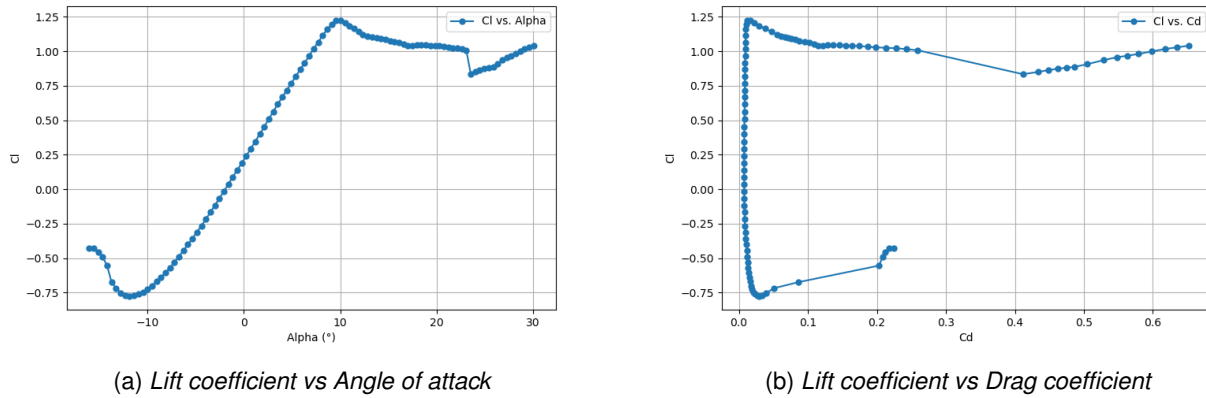
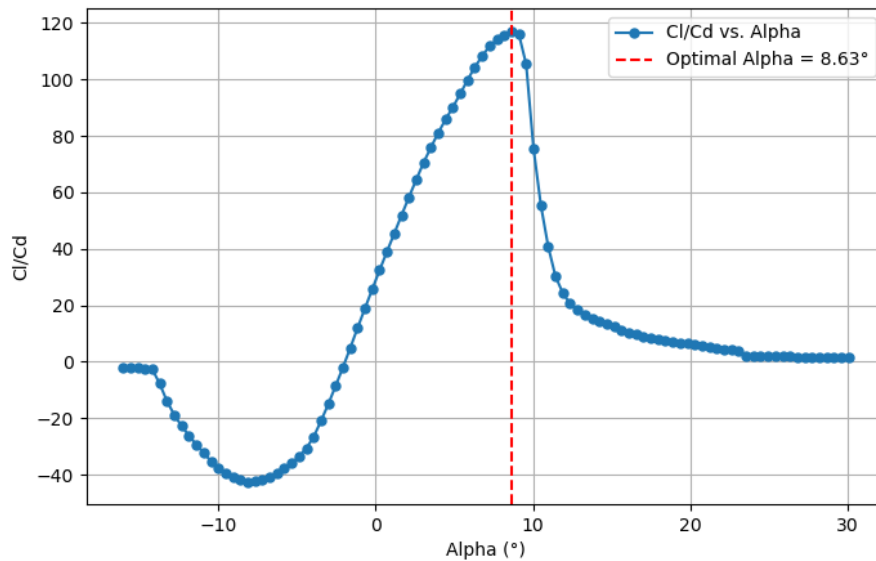


Figure 24: Airfoil lift and drag polar for the DU95W180 airfoil

To optimize performance the lift-to-drag ratio (C_l/C_d) is a very useful metric, with its peak indicating the most efficient operational point, as shown in Figure 25. Maximizing C_l/C_d is critical because it directly affects the power output and efficiency of the wind turbine; a higher value means the airfoil generates more lift for less drag, allowing the rotor to extract more energy from the wind with minimal resistance. Therefore, the airfoil should operate at the point where C_l/C_d is maximized, which for the airfoil used for this turbine corresponds to an angle of attack 8.63° . This angle of attack is approximately 1.5° from causing stall which is a useful buffer to have in reality because gusts of wind can increase the angle of attack unfavourably.



6.2.2 Lift Coefficient and Chord Distribution

The lift coefficient per unit span C_l is for a two-dimensional body and is given by Equation 20. From the equation, it can be seen that the lift coefficient is inversely proportional to the chord.

$$C_l = \frac{L}{\frac{1}{2}\rho V^2 c} \quad (20)$$

The current chord distribution of the blade is given by Equation 21 and is illustrated along the span distribution r/R in Figure 26.

$$c = \left(3 \left(1 - \frac{r}{R}\right) + 1\right) [\text{m}] \quad \text{for} \quad \frac{r}{R} > 0.2 \quad (21)$$

The variation of the chord with the span directly influences the aerodynamic efficiency of the blade. At the root, the chord length is larger, which means that the inner sections of the blade contribute more to the overall lift. Towards the tip chord length decreases, which reduces the lift coefficient. Wind turbines often compensate for this unfavourable chord length with a higher local twist or angle of attack to maintain a high lift coefficient, but here the effectiveness of lift generation at the tip is lower than at the root. Consider Figure 26 showing how the lift coefficient varies along the span of the blade.

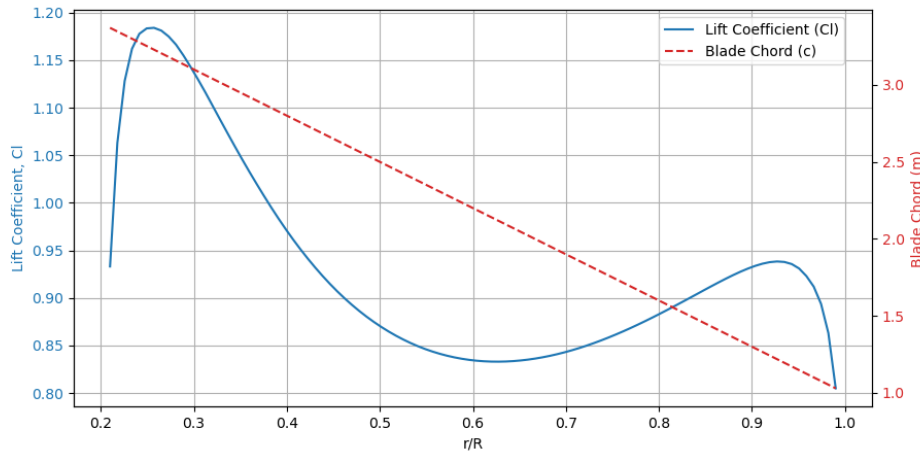


Figure 26: Lift coefficient and chord variation with span r/R .

Based on the relation described above, optimizing the chord distribution of the rotor blade is a vital part of wind turbine design. A way in which the chord can be adjusted to produce a more favourable lift distribution along the span is given in Equation 22, which is a semi-empirical formula based on experimental data [4].

$$\frac{c}{R} = \frac{16}{9} \cdot \frac{\pi}{BC_l \lambda^2} \left(\frac{r}{R}\right)^{-1} \quad (22)$$

7 Discussion and Conclusion

To summarize, this report demonstrated how the BEM code can be used to analyze the performance of a wind turbine by examining the distribution of forces along the blade span at various tip speed ratios. With the preset settings, the maximum power coefficient was found to be for a tip speed ratio of $\lambda = 9.5$. To validate the results, the impact of the tip correction factor was discussed, highlighting its significance, and a convergence study was conducted, which concluded that cosine spacing offers better computational efficiency.

The unsteady yaw model displayed unfavourable conditions that could occur in real-life scenarios, indicating that a yaw angle of 0° is optimal for maximum power production. Optimization results showed that a pitch angle of -4.25 degrees is preferred for maximizing power when a thrust coefficient of 0.75 is targeted.

Lastly, the operating points of the airfoils were analyzed, indicating that the aerodynamically optimal angle of attack is approximately 8.63° . The report also discussed the impact of stalling and briefly addressed the influence of the chord length on the lift coefficient.

The assumptions made in [section 3](#), such as uniform inflow conditions, the need for correction factors due to actuator disk assumption, and neglecting certain real-world factors like turbulence, influence the accuracy of the results. While these assumptions make the BEM tool very efficient and effective at predicting performance in comparison to CFD, they can also limit the model's capacity to predict performance accurately.

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- [4] Zaaijer, M. (2025). *Wind Turbine Design - System design and scaling*.